Question 1:

a) An estimate of μ based on the sample is about 0.997
b) the variability of my estimate is 4.71981005578034e-05
c) Using the CLT, a 95% confidence interval for μ is (0.997, 0.997)

```r
mu <- mean(red$density)

se <- sd(red$density) / sqrt(nrow(red))

ci <- mu + c(-1, 1) * 1.96 * se
```

ci	num [1:2] 0.997 0.997
mu	0.996746679174484
n	1599L
se	4.71981005578034e-05

The code and responses for questions 1a. - 1c.

d) Using bootstrap my standard error changed to 4.70277507165108e-05 and the confidence interval changed to (0.99665, 0.99684). When comparing the outputs, I can strongly say they are nearly identical, which tells me they give the same inference.

```r
set.seed(123)
B <- 10000
boot_means <- replicate(B, mean(sample(red$density, nrow(red), replace = TRUE)))

se_boot <- sd(boot_means)
ci_boot <- quantile(boot_means, c(.025, .975), names = FALSE)
```

```
> se_boot
[1] 4.702775e-05
> ci_boot
[1] 0.9966546 0.9968382
```

e) Yes, we use a normal distribution to model "density". The estimated mean is about 0.997, and the estimated standard deviation is about 0.00189. The standard errors being small means they are accurate.

```r
fit <- fitdistr(red$density * 10, "normal")
fit

mu_mle <- fit$estimate["mean"] / 10
sigma_mle <- fit$estimate["sd"] / 10
se_mle <- fit$sd / 10
```

```
> fit <- fitdistr(red$density * 10, "normal")
> View(fit)
> fit
       mean             sd
  9.9674667917   0.0188674370
 (0.0004718334) (0.0003336366)
```

Question 2:

a) An estimate of μ based on the sample is about 2.53880550343965 (2.539 rounded)
b) Yes, we use the CLT to quantify the variability of my estimate because the large sample size

```r
mu <- mean(red$`residual sugar`)

n  <- nrow(red)
se <- sd(red$`residual sugar`) / sqrt(n)
ci <- mu + c(-1, 1) * 1.96 * se
```

ci	num [1:2] 2.47 2.61
mu	2.53880550343965
n	1599L
se	0.0352592217166484

c) Using bootstrap my standard error changed to 0.0352975 and the confidence interval changed to (2.4714, 2.6086). When comparing the outputs, I can strongly say they are nearly identical to the CLT results, which tells me they give the same inference.

```r
set.seed(123)
B <- 10000
n <- nrow(red)

bmsugar <- replicate(
  B,
  mean(sample(red$`residual sugar`, n, replace = TRUE)))

se_boot <- sd(bmsugar)
ci_boot <- quantile(bmsugar, c(.025, .975), names = FALSE)
```

```
> se_boot
[1] 0.03529748
> ci_boot
[1] 2.471420 2.608568
```

d) No, we cannot use a normal distribution to model "residual sugar" because the data is highly skewed. A better option is the Gamma distribution. The estimated shape is about 6.3 and the estimated rate is about 2.48. The standard errors (0.217 and 0.089) are small, which means these estimates are accurate.

```r
fit_gm  <- fitdistr(red$`residual sugar`, densfun = "gamma")
gm_est  <- fit_gm$estimate
gm_se   <- fit_gm$sd
aic_gm  <- 2*length(gm_est) - 2*fit_gm$loglik
```

aic_gm	4402.28002600912
gm_est	Named num [1:2] 6.3 2.48
gm_se	Named num [1:2] 0.217 0.089

Question 3:

a) Using the CLT, a 95% confidence interval for pp is (0.119, 0.152).

```r
n <- nrow(red)
p <- mean(red$excellent)
se <- sqrt(p * (1 - p) / n)
ci <- p + c(-1, 1) * 1.96 * se
```

ci	num [1:2] 0.119 0.152
n	1599L
p	0.135709818636648
se	0.00856468101869562

b) Using bootstrap my standard error changed to 0.008562 and the confidence interval changed to (0.1194, 0.1526).

```r
set.seed(123)
B <- 10000
n <- nrow(red)

bmp <- replicate(B, mean(sample(red$excellent, n, replace = TRUE)))

se_boot <- sd(bmp)
ci_boot <- quantile(bmp, c(.025, .975), names = FALSE)
```

```
> se_boot
[1] 0.008561907
> ci_boot
[1] 0.1194497 0.1525954
```

c) The CLT 95% CI is (0.119, 0.152) and the bootstrap 95% CI is (0.1194, 0.1526). With this we can see that the difference is not worth our attention based on the sample size.
d) The maximum likelihood estimate of pp is 0.1357, and its standard error is 0.00856.

```r
p <- mean(red$excellent)
se <- sqrt(p * (1 - p) / n)
```

p	0.135709818636648
se	0.00856468101869562